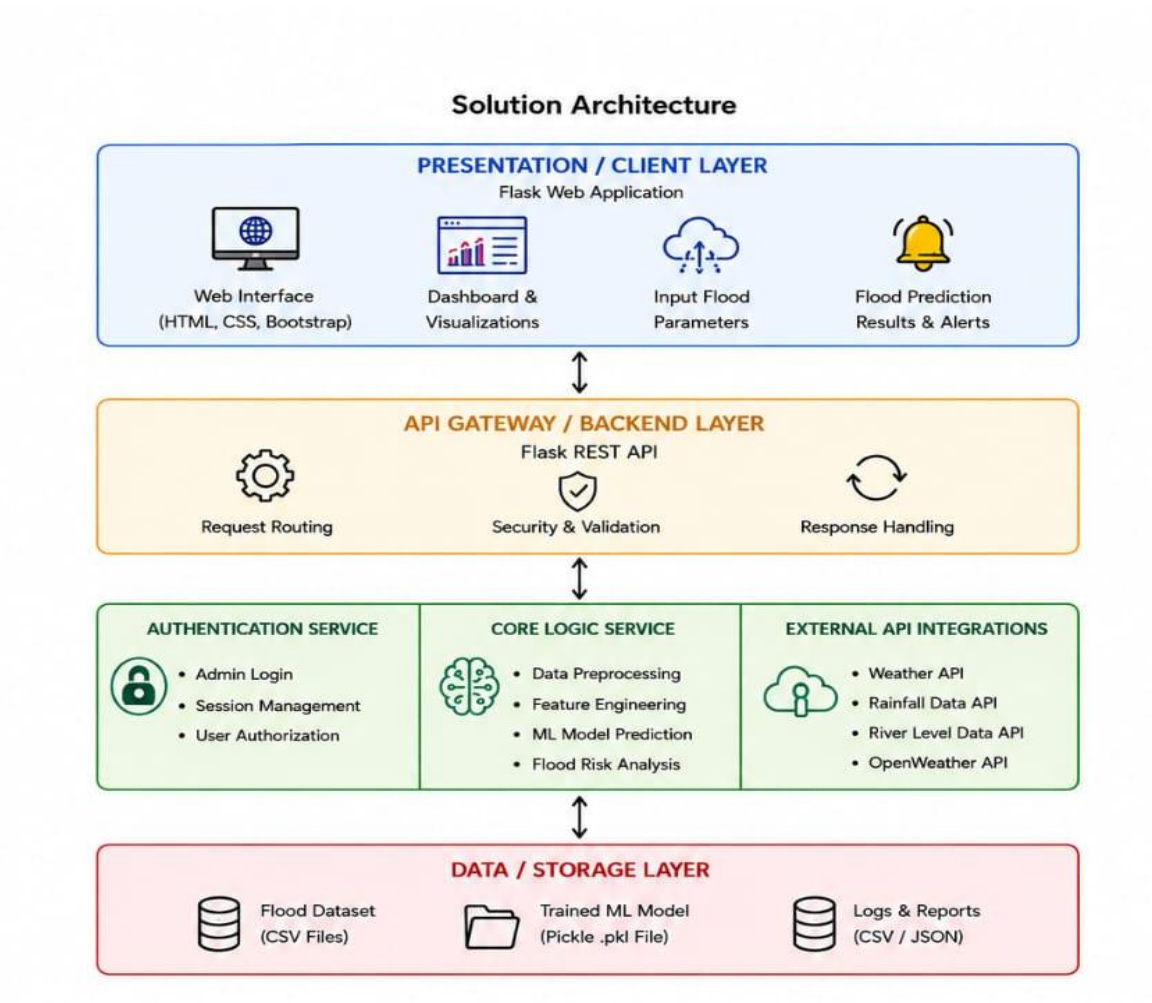


## Solution Architecture

|                      |                      |
|----------------------|----------------------|
| <b>Date</b>          | 06 July 2026         |
| <b>Project Title</b> | <b>Rising Waters</b> |
| <b>Maximum Marks</b> | 5 Marks              |

### Solution Architecture Diagram :



### Component Description Table :

| Component Name     | Description / Role in Architecture                                   | Technology used                     |
|--------------------|----------------------------------------------------------------------|-------------------------------------|
| Presentation Layer | User interface to enter flood parameters and view prediction results | Flask, HTML, CSS, Bootstrap         |
| API Gateway        | Receives user requests and sends them to the ML model                | Flask REST API, Python              |
| Authentication     | Handles admin login and user sessions                                | Flask Session                       |
| Core logic service | Data preprocessing, feature extraction, machine learning prediction  | Python, Pandas, NumPy, Scikit-learn |
| External API       | Fetches weather and rainfall information                             | Open Weather API (optional)         |
| Database / Storage | Stores dataset and trained model                                     | CSV Dataset, Pickle (.pkl)          |